

Saivenkat Jilla

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🎓 EDUCATION

University of Waterloo

Sep 2023 – Present

Bachelor of Computer Science (Honours Co-op)

GPA: 3.9/4.0

- Received Waterloo National Mathematics Scholarship (\$12,000) and President's Scholarship of Distinction (\$2,000)

🧰 EXPERIENCE

Machine Learning Researcher

Jul 2024 – Present

Alternative Protein Project

Waterloo, ON

- Designed a full workflow to extract insights from public single-cell RNA-seq data for cultivated meat applications.
- Reviewed ML methods for alternative protein research, with emphasis on cultivated meat and cell differentiation.
- Built a Python pipeline using **Scanpy**, **anndata**, **pandas**, and **scikit-learn** for quality control, normalization, feature selection, dimensionality reduction, clustering, and visualization.

Web Developer Intern

Jan 2025 – Apr 2025

Global X Investments Canada

Toronto, ON

- Built and deployed **70+** pages and posts and **10+** e-blasts (**32.4%** open rate) to support products and webinars
- Improved site ranking by cutting load times **15%** using combined API calls, lazy loading, and SEO optimizations
- Automated HTML table linking and EN/FR content syncing with **Python** and **PHP** scripts, saving **100+ hr/yr**
- Streamlined data auditing by building **MongoDB** views for an internal webpage to highlight key metrics
- Enhanced UI and customization of **WordPress** blocks by developing new and improving existing ones using **PHP**

Software Developer Intern

May 2024 – Aug 2024

Center for Work-Integrated Learning, University of Waterloo

Waterloo, ON

- Developed and tested **20+ accessible webpages** using the agile methodology within the LEARN LMS using Bootstrap and custom CSS/JS components, improving user experience and ensuring cross-platform compatibility
- Conducted QA, identifying **200+** bugs across course content, ensuring seamless user experiences and maintaining high-quality standards across course modules
- Designed and implemented a **Gantt chart using VBA** for tracking project timelines, optimizing team collaboration and task management across multiple modules

Front-End Developer

Dec 2023 – Present

Slime Scholars

Waterloo, ON

- Built key frontend components for a gamified educational platform for grade 4-12 students using **React.js**
- **Minimized latency** by **deduplicating API calls via caching data** on the front end
- Conducted thorough testing of the platform for bugs in anticipation of alpha launch

🔧 PROJECTS

🎧 devDucky | *React, Tailwind CSS, Typescript, Unsloth*

- Developed a full-stack IDE with a local code assistant, achieving **20-35%** better performance than GitHub Copilot through fine-tuned LLM models and local inference, ensuring privacy and speed
- Built a real-time dashboard for diagnostics and error tracking, improving code quality and developer experience
- Collaborated using agile methodology, starting with Figma for UI design and building the project component by component, adhering to fast iteration and feedback loops

🎧 Verbalyst | *HTML, Tailwind CSS, JavaScript, Chart.js, Google Vertex AI, AssemblyAI API*

- Developed a full-stack web application allowing users to rehearse and record speeches with both audio and camera
- Implemented feedback on the number of stutters and other speech impediments via **Google Vertex REST AI**
- Utilized **AssemblyAI's API** for accurate speech-to-text transcription on the user's uploaded recording
- Created a **robust dataset** of speech prompts, detecting impediments, and visualized progress with **Chart.js**
- Submitted to Ignition Hacks v.3 (2023), winning **Best Web Design** out of **700+** participants

⚙️ TECHNICAL SKILLS

Languages: Java, C/C++, JavaScript, TypeScript, Express, HTML/CSS, Pug, Racket, Python, Bash, VBA, LaTeX

Frameworks/Tools: React, Node, Tailwind CSS, WordPress, Bootstrap, Git, MongoDB, Visual Studio, Eclipse, Figma

Developer Tools/Libraries: Git, VS Code, Eclipse, pandas, NumPy, Matplotlib, BeautifulSoup, Scanpy, scikit-learn